Stat 145 (Multivariate Statistics)

Lesson 2.1 Inference on One Mean Vector

2025-08-31

Learning Outcomes

1. Derive the statistics for testing hypotheses on a mean vector.
2. Test hypotheses on a mean vector.

1. Introduction

One of the central messages of multivariate analysis is that p correlated variables must be analyzed jointly. We can of course do inference based on each random variable, but this could ignore the correlations of the variables at the same time lead to inflated Type I error rates. In view of the above, multivariate tests are often more powerful than univariate tests.

2. Population Variance-Covariance Matrix $\boldsymbol{\Sigma}$ Known

Given multivariate observations with p variables $\boldsymbol{y}_1, \boldsymbol{y}_2, \cdots, \boldsymbol{y}_n$,, we wish to test

$$H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0 \text{ vs } H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0.$$

More explicitly,

$$H_0 : \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix} = \begin{bmatrix} \mu_{01} \\ \mu_{02} \\ \vdots \\ \mu_{0p} \end{bmatrix} \text{ vs } H_1 : \begin{bmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{bmatrix} \neq \begin{bmatrix} \mu_{01} \\ \mu_{02} \\ \vdots \\ \mu_{0p} \end{bmatrix}$$

If we assume $\boldsymbol{y}_i \overset{iid}{\sim} N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma}), i = 1, 2, \cdots, n$, with $\boldsymbol{\Sigma}$ known, then

$$\bar{\boldsymbol{y}} \sim N_p\left(\boldsymbol{\mu}, \frac{1}{n}\boldsymbol{\Sigma}\right).$$

If H_0 is true, then

$$Z^2 = n(\bar{\boldsymbol{y}} - \boldsymbol{\mu}_0)'\boldsymbol{\Sigma}^{-1}(\bar{\boldsymbol{y}} - \boldsymbol{\mu}_0) \sim \chi^2_p$$

and we reject H_0 if $Z^2 \geq \chi^2_{\alpha,p}$.

Example 2.1.1

Consider the height (Y_1) and weight (Y_2) of $n = 20$ university students. Suppose we wish to test

$$H_0 : \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} = \begin{bmatrix} 70 \\ 170 \end{bmatrix}$$

versus

$$H_0 : \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix} \neq \begin{bmatrix} 70 \\ 170 \end{bmatrix}$$

Assume that the sample is drawn from a $N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ where

$$\boldsymbol{\Sigma} = \begin{bmatrix} 20 & 100 \\ 100 & 1000 \end{bmatrix}$$

Suppose the sample yielded the following means

$$\bar{\boldsymbol{y}} = \begin{bmatrix} 71.45 \\ 164.7 \end{bmatrix}$$

Then

$$\begin{aligned} Z^2 &= n(\bar{\boldsymbol{y}} - \boldsymbol{\mu}_0)'\boldsymbol{\Sigma}^{-1}(\bar{\boldsymbol{y}} - \boldsymbol{\mu}_0) \\ &= 20 \left(\begin{bmatrix} 71.45 \\ 164.7 \end{bmatrix} - \begin{bmatrix} 70 \\ 170 \end{bmatrix} \right)' \begin{bmatrix} 20 & 100 \\ 100 & 1000 \end{bmatrix}^{-1} \left(\begin{bmatrix} 71.45 \\ 164.7 \end{bmatrix} - \begin{bmatrix} 70 \\ 170 \end{bmatrix} \right) \\ &= 8.4026 \end{aligned}$$

The critical value at $\alpha = 0.05$, and degrees of freedom $df = p = 2$ is 5.99; and the p-value associated with $Z^2 = 8.4026$ is 0.0149761. Thus, we reject the null hypothesis.

3. Population Variance-Covariance Matrix $\boldsymbol{\Sigma}$ Unknown

Given a multivariate data $\boldsymbol{y}_1, \boldsymbol{y}_2, \cdots, \boldsymbol{y}_n$ from $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\Sigma}$ is unknown, we shall test

$$H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0 \text{ vs } H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0.$$

using the test statistic

$$T^2 = n(\bar{\boldsymbol{y}} - \boldsymbol{\mu}_0)'\boldsymbol{S}^{-1}(\bar{\boldsymbol{y}} - \boldsymbol{\mu}_0)$$

where $\boldsymbol{S}$ is the sample variance-covariance matrix.

The above test statistic is called the **Hotelling's** T^2 in honor of Harold Hotelling, a pioneer in multivariate analysis, who first obtained its sampling distribution.

Under the null hypothesis, T^2 follows $T^2_{p,n-1}$, the Hotelling's T^2 distribution with parameters p (dimension) and $v = n - 1$ (degrees of freedom). We reject H_0 if $T^2 \geq T^2_{\alpha,p,n-1}$, where $T^2_{\alpha,p,n-1}$ can be obtained from T^2 tables. It turns out that special tables of T^2 percentage points are not required for formal tests of hypotheses. This is true because

$$T^2 \sim \frac{(n-1)p}{n-p} F_{p,n-p}$$

where denotes $F_{p,n-p}$ a random variable with an F-distribution with p and $n-p$ degrees of freedom.

Thus, we reject H_0 if $T^2 \geq \frac{(n-1)p}{n-p} F_{\alpha,p,n-p}$ or if $\frac{n-p}{(n-1)p} T^2 \geq F_{\alpha,p,n-p}$.

Example 2.1.2

Perspiration from 20 healthy females was analyzed. Three components, sweat rate, sodium content, and potassium content, were measured, and the results, which we call the *sweat* data, is presented in the table below.

y1	y2	y3
3.7	48.5	9.3
5.7	65.1	8.0
3.8	47.2	10.9
3.2	53.2	12.0
3.1	55.5	9.7
4.6	36.1	7.9
2.4	24.8	14.0
7.2	33.1	7.6
6.7	47.4	8.5
5.4	54.1	11.3
3.9	36.9	12.7
4.5	58.8	12.3
3.5	27.8	9.8
4.5	40.2	8.4
1.5	13.5	10.1
8.5	56.4	7.1
4.5	71.6	8.2
6.5	52.8	10.9
4.1	44.1	11.2
5.5	40.9	9.4

Verify that

$$\bar{\boldsymbol{y}} = \begin{bmatrix} 4.640 \\ 45.400 \\ 9.965 \end{bmatrix}$$

and

$$\boldsymbol{S} = \begin{bmatrix} 2.879 & 10.010 & -1.809 \\ 10.010 & 199.788 & -5.640 \\ -1.809 & -5.640 & 3.628 \end{bmatrix}.$$

Test $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ versus $H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$, where

$$\boldsymbol{\mu}_0 = \begin{bmatrix} 4.0 \\ 50.0 \\ 10.0 \end{bmatrix}$$

The value of Hotelling's T^2 statistic is

$$
\begin{aligned}
T^2 &= n(\bar{\boldsymbol{y}} - \boldsymbol{\mu}_0)'\boldsymbol{S}^{-1}(\bar{\boldsymbol{y}} - \boldsymbol{\mu}_0) \\
&= 20 \left(\begin{bmatrix} 4.640 \\ 45.400 \\ 9.965 \end{bmatrix} - \begin{bmatrix} 4.0 \\ 50.0 \\ 10.0 \end{bmatrix} \right)' \begin{bmatrix} 2.879 & 10.010 & -1.809 \\ 10.010 & 199.788 & -5.640 \\ -1.809 & -5.640 & 3.628 \end{bmatrix}^{-1} \left(\begin{bmatrix} 4.640 \\ 45.400 \\ 9.965 \end{bmatrix} - \begin{bmatrix} 4.0 \\ 50.0 \\ 10.0 \end{bmatrix} \right) \\
&\approx 9.74
\end{aligned}
$$

```
## [1] 9.74
```

The critical value is given by

$$
\begin{aligned}
\frac{(n-1)p}{n-p} F_{\alpha,p,n-p} &= \frac{(20-1)3}{20-3} F_{0.05,3,17} \\
&= 3.353 \times 3.20 \\
&= 10.73
\end{aligned}
$$

Therefore, we fail to reject the null hypothesis at the 5% level of significance.

Learning Task

Three variables were measured (in milliequivalents per 100 g) at 10 different locations. The variables are

- y_1 = available soil calcium
- y_2 = exchangeable soil calcium
- y_3 = turnip green calcium

The data is contained in the Excel file *Calcium* (in the GitHub repo).

Test the hypothesis that $H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0$ versus $H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$, where

$$
\boldsymbol{\mu}_0 = \begin{bmatrix} 15.0 \\ 6.0 \\ 2.85 \end{bmatrix}.
$$